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Summary

One of the goals of Phys2BioMed is to develop an online platform to share raw data from nanomechanical tests on cells and tissues related to disease. Defining a universal file format for exporting of raw data from mechanical atomic force microscopy (AFM) measurements will provide the basic means for data sharing across laboratories and will include measurement parameters such as the date and time of acquisition, cantilever properties, system sensitivities, tip geometry, used instruments, calibration method, description of the sample, temperature, etc. This deliverable includes first the definition of a universal AFM force versus displacement curve being flexible to allow different types of data (simple force curves, microrheology...). The deliverable finally describes a first version of the template of the universal file format, defining the minimal parameters and data to include, and allowing an optional section to include results derived from analysis of the raw data.



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1. Introduction

1.1. Purpose of the document

The Phys2BioMed H2020-MSCA-ITN-2018 project focuses on the development of a solid scientific and technical knowledge on medical aspects of mechanics in diseases, and on advanced nanomechanical techniques for the investigation of soft biological samples of clinical relevance. This program combines research and training capacities of 9 world leading research institutes (CEA, CNRS, Friedrich-Alexander University, IBEC Barcelona, INSERM, Polish Acad. Sci., Univ. Bremen, Univ. Milano, Westfaelische Wilhelms-University Muenster), 4 public health institutions (CHU Grenoble, Jagiellonian University Medical College, San Raffaele Hospital, and Tumorothèque C2RC Lille), and 3 industrial partners (Optics 11, Vmicro, BioMeca).

One of the aims of the network is the creation of a database of raw data of mechanical tests for cross-laboratory analysis (first) and open to public (at the end of the project). This will follow the guidelines of the Open Research Data Pilot to share, deposit and maximise access to all data acquired during the project. Thus, a universal file format providing a defined structure, flexibility and usability among laboratories is essential to this aim.

At an initial state, most of the raw data will consist of force versus distance curves acquired using AFM on biological samples. The minimal information required for each force curve, including the header and raw data itself, will be defined by the establishment of the universal file format (UFF).

This document aims at defining the minimal required information of the contents and format of the data files to be shared.

1.2. Relation with the other WPs

WP1, WP2, WP4, WP5, WP6, WP9

2. Universal file format

Phys2BioMed aims at developing novel physical tools for the mechanical phenotyping of cells and tissues of clinical relevance, aiming at developing novel early-diagnostic tools. This is based on the evidence that specific diseases have particular mechanical fingerprints, therefore probing the mechanical properties of tissues, cells and extracellular matrices (ECMs) may provide hints on the progression of the disease, and possibly diagnostic evidence.

More specifically, the partners will carry on mechanical indentation experiments on selected cells, tissues, and ECMs samples of pathological relevance (listed below).

The indentation experiment typically consists in indenting a probe (for example a microsphere) against the sample, and recording the induced deformation (indentation) as a function of the applied force. By fitting a suitable mechanical model (like the Hertz equation) to the force vs indentation curves, it is possible to obtain the effective modulus of elasticity (Young's Modulus) of the sample under study.

Phys2BioMed aims at providing scientific and technological outcomes on biomechanics, and the mechanical determinants of diseases. Technology-wise, it will define standardized procedures for nanomechanical measurements, and the definition of the main features of new-generation instrumentation optimized for the mechanical phenotyping of clinical specimens. Eventually, in the longer term, Phys2BioMed aims at providing the platform and know-how to build a data bank of

mechanical fingerprints of diseases, setting the ground for the development of effective early-diagnostic tools.

One of the goals of Phys2Biomed is to provide standardized methods, including standard tools and formats for raw data processing, analysis and sharing. Thus, one of the initial tasks will be the definition and implementation of a universal file format (UFF) to share the force curves obtained during the research developed within this project. The UFF will be then publicised across the AFM community expecting it to be established as a standard in the field.

The ultimate goal of the UFF is to provide a storage of data that ascertains the reusability of the data. Thus, the UFF must integrate various kinds of data necessary to label the experiment and provide key information to analyse the data. This information should be based on a voluntary basis, so most data could be viewed as optional, but a minimal number of parameters, metadata and raw data will be mandatory to allow reusability.

2.1. Definition of the universal force curve

AFM mechanical measurements are generally carried out by the acquisition of the so-called force versus distance curves, or simply, force curves. In a typical force curve, the cantilever is moved relative to the sample using a piezoelectric element allowing the cantilever tip to contact and indent the sample and then moved back to the original position. Thus, force curves include an approach trace or segment and a retract segment. However, additional segments are possible, such as pause at constant height or constant force, low-amplitude oscillations... (Butt et al., 2005).

The deflection of the cantilever is commonly monitored by focusing a light beam (laser or superluminescent diode) on the back side of the cantilever which is then reflected and finally detected using a split photodiode. Changes in deflection are then measured as differences in the amount of light incident on the photodiode quadrants. Raw data is thus collected as a measurement of the deflection of the cantilever (in volts) as a function of the displacement of the piezoelectric element. Deflection voltage is converted into length units using the inverse of the optical lever sensitivity (invOLS) and then into force using the calibrated spring constant of the cantilever.

The basic force curve will include at least the measured cantilever deflection as a function of the piezo displacement, and it will consist of at least two segments assigned as first (1, approach) and last (N, retract) segments.

2.2. Structure of the universal file format

Apart from the raw data, the information regarding the user, instrument, calibration, etc. is also part of the force curve and is required to FAIRly share and reuse the data. Thus, the UFF will include metadata in combination with the raw data.

The heterogeneity and volume of AFM mechanical data requires a flexible file format that should be easily accessible, light and self-consistent. On the other hand, readability is an essential characteristic of FAIR open data. We have thus decided to define a main UFF based on the HDF5 architecture for data sharing and a simpler, text-based version that will provide easy exchanges.

HDF5 creates a container that include its structure organization; thus it is self-consistent and the structure of the container can be read from scratch (however, not as ASCII text). To generate HDF5 files, it is necessary to use the HDF5 libraries that are available in several programming environment (C, Fortran, Python). The downside of this concept is the requirement to develop a specific tool for reading/writing the data (to the opposite of a simple text editor for instance). Thus, the ASCII-based version will provide further exchangeability. In any case, a conversion tool will be developed to export AFM data from different manufacturers into both UFF formats.



2.2.1. HDF5-UFF structure

For the HDF5-UFF, the structure has to be defined first. We thus first describe the general HDF5 architecture.

HDF5 files are composed of three main elements:

- **Groups:** These are the containers used to organize the contents of the file. They can be compared to directories/folders.
- **Datasets:** They are homogeneous collections of data elements, with a fixed datatype and (hyper)rectangular shape.
- **Attributes:** Small pieces of data directly attached to Groups and Datasets used to store metadata.

Thus, in the HDF5-UFF, the metadata will be in the form of attributes associated to the corresponding group containing the particular dataset (deflection, height...) (Fig. 1). Given the flexibility of HDF5 format, other datasets, such as calibration curves, thermals or analyzed data could be included.

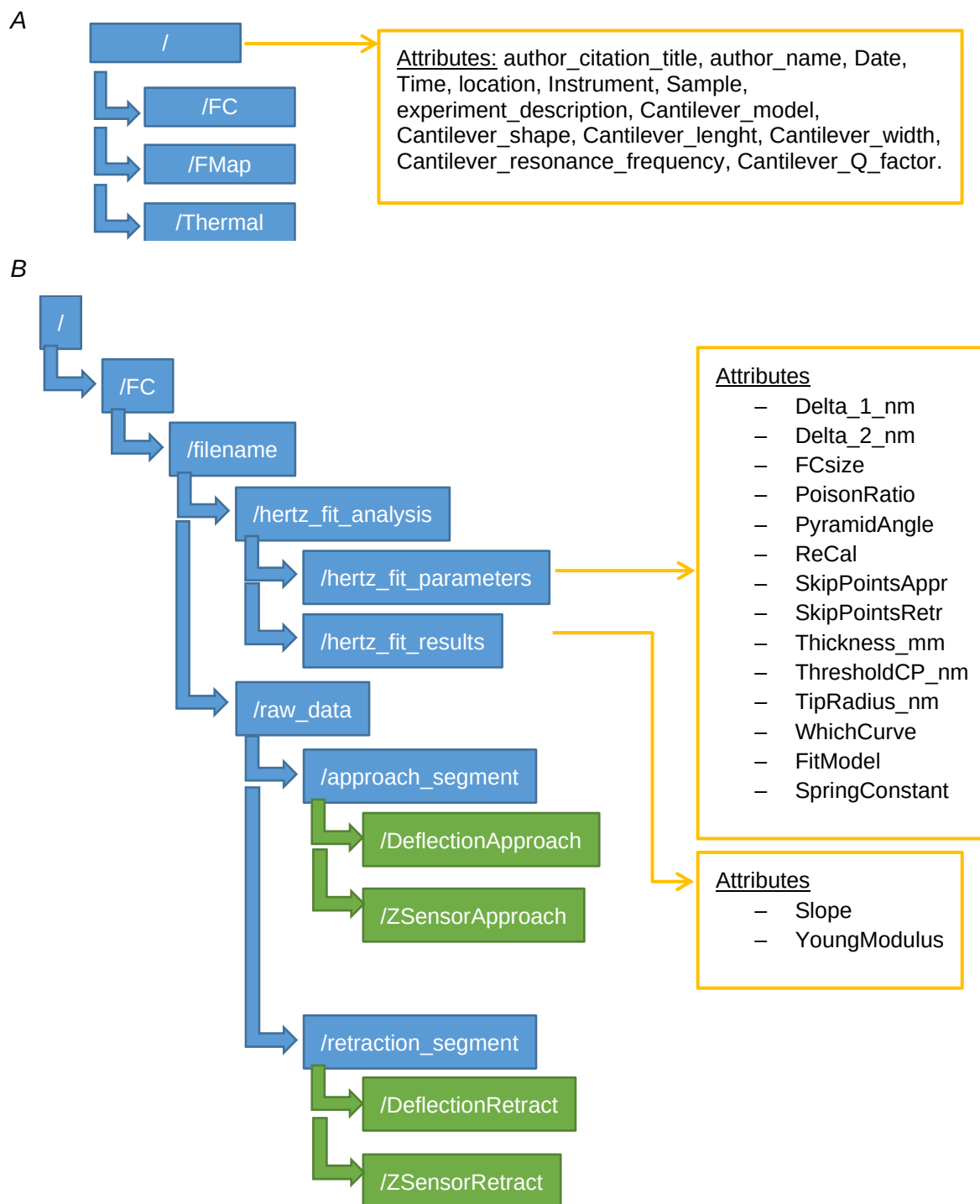


Figure 1. HDF5 Universal file format. **A** Each type of data will be contained in a group (FC, force curve; FMap, force map...). The metadata will be added in form of attributes to each group and will provide the minimal information about the force curves regarding the used system, experimental conditions.... Each group of data will contain a group defined by the filename with the raw data for each segment **B** Example of the structure of the data for a force curve (FC)

2.2.2. ASCII-UFF structure

For the ASCII-UFF, the structure of the file will combine a header coded followed by measured data coded in ASCII (Fig. 2). Both header parameters and data will be provided in SI units.



Figure 2. Universal file format. A header in ASCII format will provide the minimal information about the force curves regarding the used system, experimental conditions.... The header will be followed by the force curve data also in ASCII format.

2.2.3. Metadata (Attributes or Header)

Regardless of the HDF5 or ASCII based format, several parameters will be required as listed below. Mandatory parameters are in a), c), d), e) and f):

- a) File information: provide a unique identifier to this file, as well as the date and version
- b) Authors information: Institution address, author's name, author's associated publication of this data
- c) Biological system: describe the entities used in the experiment (PROBABLY the most complicated term). Define molecule names, cell names, species, origin of molecules...
- d) Experimental data: How the data was performed in terms of metrology parameters (Air, liquid, Temp, Buffer name and composition, pH, ...)
- e) Tip information: Provider, parameters (length, k, coating,...), should we put nominal values or measured values (likely both)
- f) AFM information: ramp size, ramp speed, number of segments and description (approach, contact, retract, pause), nb of points, sampling rate, def_sensitivity, z_sensitivity,...
- g) Analysis parameters: Model used (paraboloid, regular pyramidal, conical, spherical,...) and tip geometry (angle, radius...), Poisson ration, fitting methods, all kinds of corrections (baseline, bottom effect,...),
- h) Result section: Young modulus, apparent modulus, stiffness, Max indentation, contact point distance, Max force,...

HDF5-UFF

Metadata will be provided in form of attributes to each group containing the datasets.

ASCII-UFF

The header in ASCII format will have a fixed extension of 81920 bytes.

2.2.4. Data

The minimal information required for the data section will be “deflection” signal in Volts.

The piezo movement could be, if not directly measured and/or stored, calculated from the header parameters.

The data section in binary format will not have a fixed extension and will depend on the length of the acquired data. The use of binary format for the data will minimize considerably the size of the files.

2.3. File template

A possible template for the attributes or header metadata and the data itself is shown below. Its improvement and completeness will be updated along the whole project. The final mandatory and optional parameters are to be determined. Asterisks * stand for the minimal mandatory parameters:

```
#Header universal force curve file
DATA_<file_code>
ENTRY
  _UFF_version
  _date
  _file_version
  _text_length
  ...
AUTHOR
  _author_name
  _author_citation_title
  _location
  ...
ENTITY
  _entity_sample_name
  _entity_sample_species
  ...

EXPERIMENTAL
*Instrument:
*Sample: cell X, protein X...

CANTILEVER
Cantilever model: MLCT-E, PFQNM-LC...
*Spring constant:
Calibration method: nominal, Sader, GCI, vibrometer...
Cantilever shape: V-shape, rectangular...
Cantilever length:
Cantilever width:
Cantilever resonance frequency:
Cantilever Q factor:
...

TIP
```

*Tip geometry: four-sided regular pyramid
*Tip parameters: angle 35

...

RECORDING

xposition: -4.7500e-06
yposition: -2.2500e-06
*piezo z sensitivity: 14e-09
*inVOLS: 22e-9
*No_segments: 4
*Segment 1 type: approach
*Segment 1 length: 4e-6
*Segment 1 no points: 1024
*Segment 1 sampling rate: 1e5
Segment 1 velocity: 1e-6
Segment 1 force set point: (if any)

*Segment 2 type: constant height
*Segment 2 no points: 100
*Segment 2 sampling rate: 1e5
Segment 2 velocity: 1e-6

...

*Segment N type: retract
*Segment N length: 4e-6
*Segment N no points: 1024
*Segment N sampling rate: 1e5
Segment N velocity: -1e-6

...

NOTES

ANALYSIS

Data processing: tilt correction using a 1-polynomial fitted on segment 1 points 1 to 100

...

Data scaling deflection: converted to force using inVOLS and kcanti
Data scaling zpiezo: converted to length using zsens*

...

Segment 1 analysis method: fit contact model
Segment 1 Young's modulus:
Segment 1 model: paraboloid
Segment 1 model radius:
Segment 2 analysis method: fit exponential decay
Segment 1 contact model: paraboloid
Segment 2 relaxation time:

...

Segment N Young's modulus:

NOTES



*Datatype "Unit"

Time "s", Height "m", ZSensor"m", ZVoltage "V", Deflection "m", Lateral
Deflection "m", TappingError "Hz", PeakForce Error "N"

==== ===== {END OF TEXT DATA}

*(Data in binary (HDF5-UFF) or text (ASCII-UFF) format {X1 Y1 Z1\r X2 Y2
Z2\r...})

3. Towards establishment of the universal file format

For a file format to be universal and used widely, it is essential to disseminate it within the community of users and to provide tools to adopt it easily. In this section, some of the intended tasks to that end are defined.

3.1. Distribution across the AFM community

The most established tool to communicate scientific and technological advancements is the publication of the results or developments in peer-reviewed journal. The current trend of going towards an open science involves sharing the results of scientific research in the form of raw data, allowing reusability. The most efficient dissemination of a computer tool is to provide accurate and complete description of its format in an open-access system.

There are currently some options to share data on repositories and this project aims at using them, first internally and later opening them to the community. We aim at disseminating UFF as a publication in a peer-reviewed journal, for example, "[Scientific data](#)", "[Dataset papers in Science](#)", "[Data Science Journal](#)" or the new H2020 Open Research Europe publication platform.

Moreover, once the UFF is defined, dissemination across the AFM community will also be carried out by directly contacting AFM companies and providing help to implement the UFF export option in their AFM software. Feedback from AFM companies as well as users will help improve the format and thus its dissemination.

3.2. Development of tools for UFF usability

Furthermore, establishment of the UFF will be easier if it is complemented by appropriate open software tools to export AFM company files into the UFF and for import, processing and analysis of the data to extract mechanical information of the sample.

Phys2BioMed will also develop and share such tools. Collaboration with other open software for AFM data analysis is already ongoing and will continue during and beyond the Phys2BioMed project lifetime. Indeed, part of these tools are already being implemented and are part of WP, in particular deliverable 4.2.

There are several practical steps to follow for reaching our goal that is to define the UFF for force-distance curves for FAIR sharing and reusing. The expected roadmap will be as follows

- a) Define the architecture of the UFF
- b) Create a library that read manufacturer AFM files and convert into the UFF

-
- c) Create a library that read the UFF and convert to whatever (like simpler text files which are easily manipulated by home-made software)
 - d) Create a graphical interface that link the above tasks
 - e) Create a graphical interface used to fill-up metadata required for the UFF

4. References

Butt, H.J., Cappella, B., and Kappl, M. (2005). Force measurements with the atomic force microscope: Technique, interpretation and applications. *Surf. Sci. Rep.* 59, 1–152.

